$\therefore p \to (q \vee r)$ is True.

$\therefore$ If $x, y \in R$ and $(x+y) \geq 2$, then $x \geq 1$ or ~~y≥1~~ $y \geq 1$ □.

18) Prove that if m and n are integers and mn is even, then m is even or n is even.

$\boxed{\text{Soln:}}$ p: m, n are integers, mn is even q: m is even integer
r: n is even integer

We shall prove that $p \to (q \vee r)$ is True by contraposition.

$\therefore$ We shall prove $\neg(q \vee r) \to \neg p$ is True.

Let, $\neg(q \vee r)$ be True. By De-Morgan's Law, $(\neg q \wedge \neg r)$ is True.

By defn. of AND, $\neg q$ is True and $\neg r$ is True.

$\neg q$: m is not even, i.e. m is odd (Defn. 1)

$\neg r$: n is not even, i.e. n is odd (Defn. 1)

By Defn 1, $\exists$ integers K_1 and K_2 such that $m = 2K_1 + 1$ and $n = 2K_2 + 1$

$mn = (2K_1+1)(2K_2+1) = 4K_1K_2 + 2K_1 + 2K_2 + 1 = 2(2K_1K_2 + K_1 + K_2) + 1$

It's easy to see that $(2K_1K_2 + K_1 + K_2)$ is an integer.

$\therefore$ By Defn. 1, mn is odd. $\therefore$ mn is not even is True

$\therefore$ mn is even is False $\therefore \neg p$ is True

$\therefore$ mn is even is False $\therefore \neg p$ is True.

By defn. of conditional, $\neg(q \vee r) \to \neg p$ is True.

$\boxed{\therefore p \to (q \vee r) \text{ is True.}}$

So, if m and n are integers and mn is even, then m is even or n is even. □

19) Show that if n is an integer and n^3+5 is odd, then n is even using
a) a proof by contraposition b) a proof by contradiction.